

One-Day Hands-On Workshop on Cloud Computing With AWS

About The Program

Cloud Computing Has Become the Backbone of Modern Digital Applications, Enabling Scalable, Secure, And Cost-Effective Deployment of Software and Machine Learning Solutions. Amazon Web Services (AWS), As the Leading Cloud Service Provider, Offers A Wide Range of Services That Support Real-World Application Development and Deployment Across Industries.

This One-Day Hands-On Workshop On “Cloud Computing With AWS” Is Designed to Provide Participants with A Strong Foundation in Cloud Computing Concepts and Practical Exposure to AWS Core Services. The Program Begins with An Introduction to Cloud Fundamentals and AWS Global Infrastructure, Followed by An In-Depth Exploration of Essential Services Such as Amazon EC2, S3, And IAM.

A Major Highlight of The Program Is the End-To-End Deployment of a Machine Learning Application on An AWS EC2 Instance. Participants Will Gain Hands-On Experience in Launching and Configuring an EC2 Instance, Connecting to It Securely, Installing Required Software, And Deploying An ML-Based Web Application Using Python and Flask. The Workshop Also Covers Best Practices Related to Security, Monitoring, And Cost Management In AWS.

By the End of The Program, Participants Will Be Able to Confidently Work with AWS Cloud Services and Deploy Real-World ML Applications in A Cloud Environment. This Workshop Is Ideal for Students, Faculty Members, And Professionals Who Are Interested in Cloud Computing, Machine Learning Deployment, And Building Industry-Ready Skills.

Tools:



Introduction to Cloud Computing & AWS

Session	Topic	Content
09:30 AM – 10:30 AM	Introduction To Cloud Computing & AWS	Cloud Computing Basics (IaaS, PaaS, SaaS), Advantages Of Cloud, AWS Overview, Regions & Availability Zones, Overview Of EC2, S3, IAM
10:30 AM – 11:30 AM	AWS Compute & Storage Services	EC2 Fundamentals, Instance Types, AMI, Key Pairs, Security Groups, S3 Buckets, IAM Basics And Security Best Practices
11:30 AM – 11:45 AM	Tea Break	
11:45 AM – 01:15 PM	Launching & Accessing EC2 Instance	Launch Ubuntu EC2 Instance, Configure Security Groups, SSH Access (Windows/Linux/Mac), Basic Linux Commands, Install Python & Pip
01:15 PM – 02:00 PM	Lunch Break	
02:00 PM – 03:30 PM	ML Application Setup On EC2	Machine Learning Project Overview, Flask Architecture, Uploading Model Files (.Pkl), Installing Dependencies (Flask, NumPy, Pandas, Scikit-Learn), Testing ML App
03:30 PM – 04:15 PM	Deploying ML App On AWS EC2	Running Flask App On EC2, Gunicorn Overview, Port Configuration (5000/80), Accessing App Via Public IP
04:15 PM – 04:30 PM	Monitoring, Cost & Wrap-Up	CloudWatch Overview, AWS Cost Management, Free Tier Limits, Stopping EC2 Instances, Q&A And Conclusion